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Teaching Machines to “Feel” Tweets

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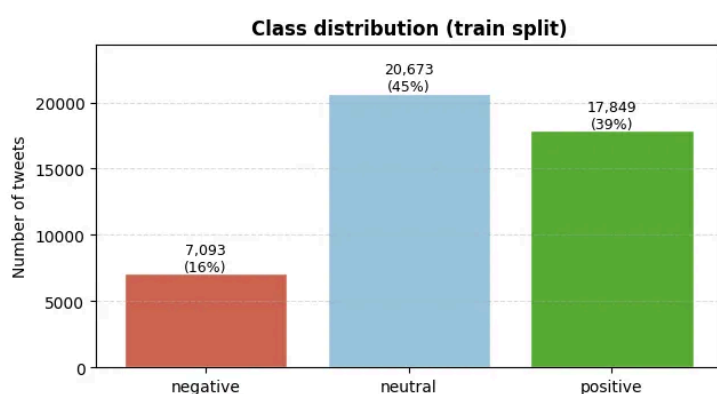
In every minute, about half a million tweets are published. These tweets span opinions on the news, elections, products, public figures, and personal crises which pour onto our feeds! Despite their conciseness, they can be ambiguous, making them difficult to interpret correctly. Twitter language, as I like to call it, is terse, heavily abbreviated, loaded with slang and jargon, hashtags, and sarcasm. A sentence that appears positive in a newspaper might be ironic on X! For instance, you might see in a newspaper: “A Prominent Tech Executive Sparks Historical Debate on Political Ideologies”. BUT A TWEET RESPONDING TO THE SAME EVENT MIGHT SAY:



Both communicate sentiment, but one does it literally, while the other carries an air of ambiguity. There is always a gap between lexical meaning and social meaning, let's say. On social media like X, words alone are insufficient. Context, culture, tone and sentiments carry far greater importance. For researchers, businesses, brands and, say, policymakers, accurately classifying the sentiment of tweet data is tremendously valuable, but genuinely arduous.

The Data

Before teaching a machine to “feel”, you must first define what those feelings are. Luckily, there is a TweetEval sentiment dataset, sourced from Hugging Face’s datasets library. It provides the empirical foundation, since it arrives pre-split and pre-labelled across three sentiment classes, which makes life less ambiguous: positive, negative and neutral feelings. Its class distribution is interesting. “Neutral” tweets dominate, carrying 45% of the training set, followed by “positive” tweets at 39%, with “negative” tweets representing a mere 16%. In many ways, I would say it depicts real world social media behaviour. Users are usually being factual, ambiguous and if they are being careful, observational.



Sample tweets by sentiment

Label	Tweet (truncated)
negative	"#CNET - just bought my 1st iPad, iPad3, feeling real burned, mad...
negative	"Mother's Day is coming like tomorrow or is it today? idk, but i ...
negative	Can't believe I left my charger at Tia's. I have just under 50% u...
neutral	@user @user @user I think this is the motive of the Yakub's layw...
neutral	@user From this performance it is obvious why David Miliband come...
neutral	@user @user We are actually playing on Saturdayu002c January 19...
positive	Kanye West was honored in a big way during Sunday night's MTV Vid...
positive	Have you heard about Amazon Prime day? Don't miss it - July 15th!...
positive	Houston peeps: Go see Rumours (Fleetwood Mac cover band from Los ...

Distribution of sentiment classes in the TweetEval Twitter dataset (train split). Sampled tweets with corresponding labels

Enter the Transformers

Ok so once the scale and complexity of tweet data is realised, you can see how impossible it is to analyse it conventionally. We therefore bring in some indispensable heavy lifters, transformer models! The approach here belongs to a process called Transfer Learning. Instead of training a model from scratch, you start with a model that has already learnt vast amounts of linguistic structure, if you like. You can then adapt that model to a specialised task, which in our case is tweet sentiment classification. We explore some interesting transformer architectures below.



Not the Transformers! source: [Aditya Vyas](#)

The BERT Family

BERT

We select BERT (Bidirectional Encoder Representations from Transformers) as our primary baseline model. The model reads text in both directions simultaneously. Rather than processing a sentence left-to-right as older RNN-based models did, BERT uses multi-head self-attention across all tokens at once, allowing each word to “see” every other word before its meaning is fixed. Pre-trained on billions of words from Wikipedia and BooksCorpus, it learns rich contextual representations that transfer well to downstream classification tasks such as ours.

DistilBERT

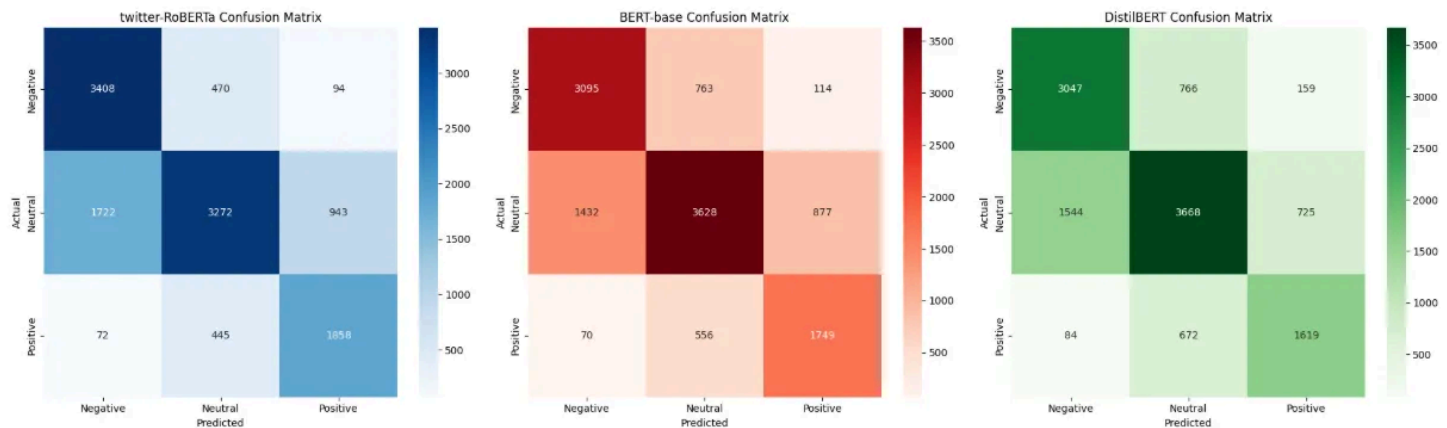
DistilBERT is our second baseline. It’s akin to a smaller “student” version of BERT, trained to imitate the behaviour of a larger “teacher” model. It retains most of BERT’s characteristics despite being smaller, faster, lighter and cheaper.

RoBERTa

We also use RoBERTa, which is a trained evolution of BERT. It also retains the same encoder architecture but was trained on much larger datasets, with longer training runs and larger batch sizes. One version of the model, called twitter-RoBERTa (twitter-RoBERTa-base), was pre-trained on over 58 million tweets before being subsequently fine-tuned on the TweetEval dataset. This makes it a genuine domain-

native model. It has already encountered emojis, hashtags, informal abbreviations, and “all-caps rage” before. Its vocabulary and contextual representations are shaped by the same kind of text it is being asked to classify.

Findings



Confusion matrices of the fine-tuned models

The insights emerged from the visuals of confusion matrices as well as the values of the models’ macro recall and macro F1 scores. Twitter-RoBERTa draws cleaner separations between “negative” and “positive” sentiments, showing the lowest misclassification rates in the off-diagonal areas across board. However, DistilBERT does the best job differentiating between “negative” and “neutral” sentiments compared to the others. This is actually the crucial piece in applied settings, because a sentiment monitoring system that consistently misreads frustration as indifference would give systematically distorted signals for analysts, businesses or brands.

The Tokenisation Gap

The performance gap between BERT and twitter-RoBERTa can be traced to a fundamental tokenisation mismatch. BERT uses WordPiece tokenisation, designed for formal text. When it encounters hashtags, like *#CNET*, an emoji, or an abbreviation like *idk*, it’s likely to split these into sub-word fragments that carry no coherent meaning in its pre-trained vocabulary, treating them as noise rather than signal.



Twitter-RoBERTa, on the other hand, was already trained on millions of tweets using Byte-Pair Encoding. It has already encountered these patterns several times during pre-training. This prior exposure means less representational loss at the tokenisation stage, which compounds across all 12 encoder layers into meaningfully better classification. In essence, the model is processing the culture and context embedded in Twitter Language, which is fascinating.

Conclusion

In summary, this study reinforced a principle that sounds intuitive, but is still worth empirically demonstrating: *pre-trained models within a particular domain are essential*. If you are attempting to classify language between two polarised sentiments on X, a model trained extensively on Twitter discourse will outperform a general-purpose language model precisely where the errors matter most. When classifying text written on X, a model trained extensively on Twitter discourse performs measurably better. It has learned the patterns and social contexts embedded in the environment the sentiments are expressed in. On social media platforms, meaning is rarely carried by words alone. However, if you are keen on differentiating neutral from negative discourse for business use cases, such as brand opinion/preference, power your AI system with DistilBERT.

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